

Vikash Kumawat

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SUMMARY

BCA (Data Science) student and Data Science Intern at Regex Software with 9+ months of hands-on experience building scalable backend systems, end-to-end ETL/ML pipelines, and production-ready Flask APIs. Proficient in Python, SQL, Flask, Pandas, and scikit-learn. Experienced in relational database design, REST API development, Docker containerization, and interactive data visualization. Seeking a fresher role in Data Analytics, Data Engineering, or Backend Development.

TECHNICAL SKILLS

Languages: Python, SQL, HTML5, CSS3, JavaScript

Backend / Web: Flask, REST API Design, CRUD Operations, Component-Based Architecture

Databases: MySQL, Relational Database Design, INNER/LEFT/RIGHT JOINS, Foreign Keys, ETL Pipelines, GROUP BY / AVG / COUNT

Data & Analysis: Pandas, NumPy, Matplotlib, Seaborn, Plotly, EDA, Data Preprocessing, Feature Engineering

Machine Learning: scikit-learn, Linear Regression, Logistic Regression, Decision Tree, Random Forest, SVM, K-Nearest Neighbors (KNN), K-Means Clustering, DBSCAN, StandardScaler, MinMaxScaler, Label Encoding, SMOTE

Tools & Platforms: Git, GitHub, Docker, VS Code, Jupyter Notebook, Google Colab, Prompt Engineering

Operating Systems: Linux (Ubuntu / Bash), Windows

Concepts: OOP, DBMS, System Design, Supervised Learning

EXPERIENCE

REGex Software Services

Jaipur, Rajasthan

Data Science Intern

Sep 2025 – Present (9+ months)

- Engineered ML-ready datasets from raw CSVs using Pandas — performing null imputation, label encoding, and categorical feature mapping — delivering clean pipelines for downstream machine learning workflows.
- Conducted end-to-end EDA on COVID-19 and Titanic datasets using Pandas and NumPy, identifying demographic and geographic patterns through univariate and bivariate analysis across 1,000+ records.
- Developed interactive dashboards using Matplotlib, Seaborn, and Plotly — heatmaps, distribution plots, and donut charts — to present actionable data insights to stakeholders.
- Built end-to-end ML pipelines in Python (scikit-learn) covering data ingestion, preprocessing, model training, and Pickle serialization — reducing manual pipeline setup time by ~40%.
- Developed and deployed Flask REST APIs serving regression and classification models for insurance charge prediction and subscriber churn prediction.
- Dockerized a Flask + MySQL application with custom container networking — resolved localhost-to-container hostname issues for consistent, production-ready multi-container deployment.
- Designed prompt-engineering workflows to automate repetitive data reporting tasks, reducing manual effort by 60% through structured AI instruction pipelines.

PROJECTS

AmazonScout — AI Product Research Agent | Python · Flask · Groq/LLaMA 3.3 70B · ScraperAPI · RainforestAPI 2025

- Built a full-stack AI-powered Amazon product research agent that scrapes live product data and feeds it to a Groq-hosted LLaMA 3.3 70B model to generate structured research summaries.
- Engineered a Flask backend with modular API routes orchestrating multi-source data fetching, LLM prompt construction, and JSON response handling in a single pipeline.
- Integrated prompt engineering techniques to guide LLaMA 3.3 70B in producing consistent structured outputs including price comparisons, review sentiment, and product recommendations.

Supervised ML Models — Churn & Insurance Prediction | Python · scikit-learn · Flask · Pandas · SMOTE 2025

- Built a Customer Churn Prediction API using Random Forest and Decision Tree classifiers on Churn_Modelling dataset (10,000+ records) — handled class imbalance with SMOTE, achieving significant F1-score improvement.
- Developed an Insurance Charge Prediction API using Linear Regression on structured medical cost data — applied feature engineering, StandardScaler normalization, and served real-time predictions via Flask REST endpoint.
- Trained and compared Logistic Regression, Decision Tree, Random Forest, and SVM models, evaluated using accuracy, precision, recall, and F1-score metrics.

DriveX — Car Booking & Feedback Web App | Python · Flask · MySQL · Docker

2025

- Developed a full-stack car booking web application handling end-to-end booking flows, user registration/login, and structured feedback collection.
- Designed a normalized MySQL relational schema with foreign key constraints linking users, bookings, and feedback tables.
- Containerized the application using Docker with multi-container networking (Flask + MySQL), resolving localhost-to-container hostname issues for production-ready deployment.

- Mitti Cafe — Full-Stack Web Application | Python · Flask · JavaScript · HTML5 · CSS32025
- Architected a multi-page cafe web application with a scalable flat-file structure and component-based integration system spanning 6 pages.
 - Engineered a reusable `components.js` module that auto-injects navigation and footer elements across all pages at runtime — eliminating code duplication.
 - Implemented accessibility-first design with skip-links, semantic HTML5 landmark elements, and responsive CSS3 layouts.
- Data Analytics & EDA Pipeline | Python · Pandas · Plotly · scikit-learn2025
- Built end-to-end ETL pipelines to ingest, clean, and transform raw CSV datasets — handling nulls, encoding categoricals, and removing outliers.
 - Performed EDA on Employee Attrition (1,400+ records), COVID-19, and Titanic datasets — uncovering patterns via univariate and bivariate analysis.
- Booking Management System | Python · Flask · MySQL · Docker2025
- Designed a normalized relational MySQL database schema with multiple tables linked via foreign key constraints to manage bookings, users, and resources at scale.
 - Built RESTful backend API endpoints using Flask with full CRUD operations and optimized SQL JOIN queries for cross-table data retrieval.

EDUCATION

- Commerce College JaipurJaipur, Rajasthan
- Bachelor of Computer Applications (BCA) — Data Science2023 – 2026 (Expected)
- Relevant Coursework: Data Structures & Algorithms, DBMS, Python, Machine Learning, Web Technologies, Statistics
- Surendra Public SchoolJaipur, Rajasthan
- Higher Secondary Certificate (12th) — CommerceCompleted 2023

CERTIFICATIONS & ACHIEVEMENTS

- SQL (Basic) Certification — HackerRank (2024)
- SQL (Advanced) Certification — HackerRank (2024)
- AI Tools Workshop — Artificial Intelligence & Automation (2024)
- Google Data Analytics Certificate — Google / Coursera (In Progress, 2025)
- Python, Pandas & Data Visualization — Kaggle (In Progress, 2025)
- Public Learning: Documenting a structured day-by-day Machine Learning course on GitHub and LinkedIn — building a consistent, verifiable learning record.